

Reinforcement Learning - Sutton and Barto

Book Club Highlights

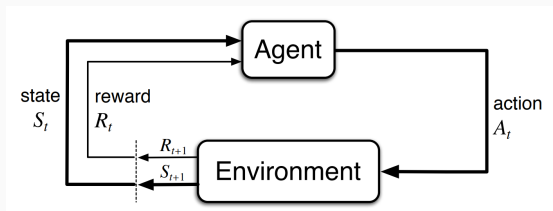
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Chapter 3

Finite Markov Decision Processes

Agent-Environment Interface



agent learner and decision maker

environment everything outside the agent

$S_t \in \mathcal{S}$ state of the environment at time t

$A_t \in \mathcal{A}(s)$ action at time t

$R_t \in \mathcal{R} \subset \mathbb{R}$ reward received as a consequence of action A_{t-1}

Markov Decision Processes (MDP)

Markov Property

The probability of each possible value for S_t and R_t is completely determined by the preceding state and action S_{t-1} and A_{t-1} .

Dynamics Function

The function $p : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$ is defined as

$$p(s', r \mid s, a) \doteq \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

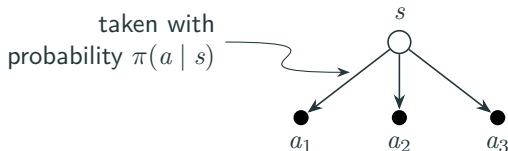
and completely defines the dynamics of the MDP.

Policies and Value Functions

Policy

Mapping from states to probability of selecting each possible action.

$$\pi : \mathcal{S} \mapsto \{\text{probabilities over } \mathcal{A}(s)\}$$

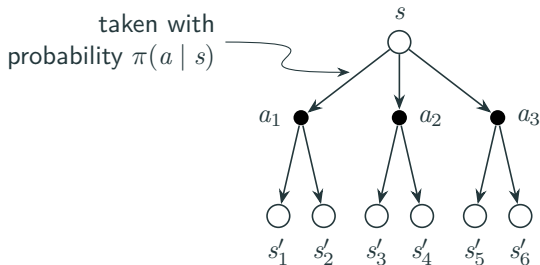


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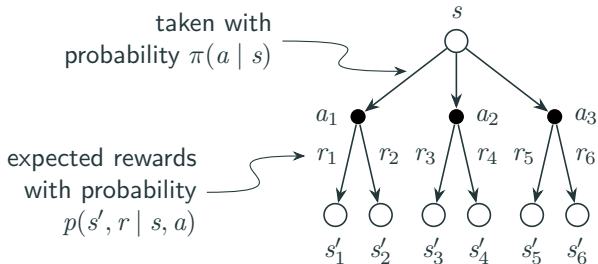


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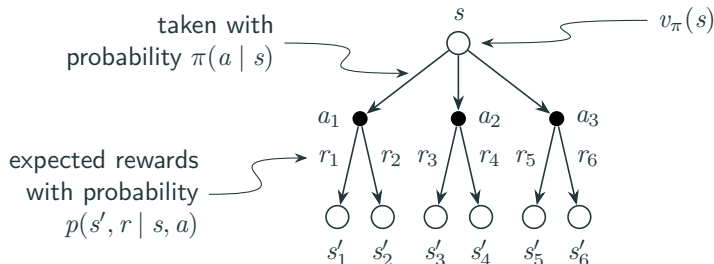


Policies and Value Functions

State value function

Expected return when starting in a given state s and following policy π thereafter.

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$$

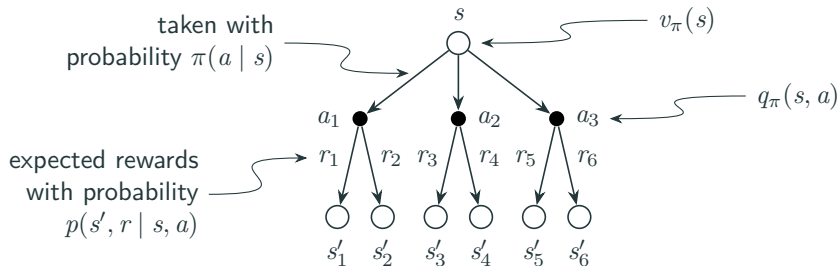


Policies and Value Functions

Action value function

Expected return when starting in a given state s , taking an action a , then following policy π thereafter.

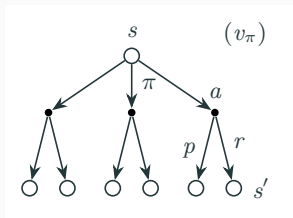
$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$$



Policies and Value Functions

Bellman Equations

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) [r + \gamma v_{\pi}(s')]$$



Policies and Value Functions

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$$q_{\pi}(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) \left[r + \gamma \sum_{a' \in \mathcal{A}(s')} \pi(a' | s') q_{\pi}(s', a') \right]$$

